ÍCARO SANTOS DE OLIVEIRA

PhD Student in Computer Science · Software Quality & Test Engineering

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PROFILE

PhD student in Computer Science at the State University of Ceará (UECE), researching **testing and quality assurance of LLM-based systems with Retrieval-Augmented Generation (RAG)**. Holds an MSc in Computer Science (UECE) on testing Big Data systems and data quality. ISTQB® certified, with over six years of hands-on experience in E2E, API, load and security test automation integrated into CI/CD pipelines, across mission-critical government and enterprise projects.

EXPERIENCE

Systems Analyst — Software Quality

Ceará Water and Sewage Company (CAGECE)

2025 – Present Fortaleza, Brazil

- **Protheus ERP quality team:** led the transition from ad-hoc manual validation to a multi-layer continuous quality programme — UI automation with TIR and API testing with Playwright, Allure reporting and Jenkins pipeline-as-code. Outcomes: 30+ critical payroll routines and 1,300+ automated API tests.
- **AI team:** designed the test architecture for the organisation's RAG assistant, based on hybrid oracles — cosine semantic similarity (spaCy, 0.85 threshold) combined with deterministic guardrails. 53 test cases reaching a 96.22% pass rate over three months of CI, catching real regressions after regulatory updates.
- **Quality governance:** Gitflow for test assets, a dedicated Quality Board and a quarantine lane to manage flakiness and test technical debt.

QA Lead / QA Analyst

GREat — Federal University of Ceará (UFC)

2022 – Jun. 2025 Fortaleza, Brazil (Hybrid)

- Led the QA team of the **Antonieta de Barros** project (FNDE): set up E2E automation with Playwright and API testing with Pytest, integrated into GitLab CI/CD pipelines.
- Testing process for the **Big Data Fortaleza** platform (Iplanfor): functional and exploratory testing, unit and integration testing (JUnit 5, Testcontainers/Docker), API testing (Postman), load testing (JMeter, Locust) and security testing (OWASP ZAP).
- Additional work as a junior full stack developer: React on the front end, Spring with PostgreSQL on the back end, and Go/Python microservices on AWS (EC2).

Development Intern

Etice — Ceará State Information Technology Company

2023 – 2024 Fortaleza, Brazil

- Web development with Django and PostgreSQL, Google-based authentication and analytical dashboards built with pandas and Plotly.

EDUCATION

PhD in Computer Science

State University of Ceará (UECE)

2026 – 2029 (ongoing)

- Thesis: testing and quality assurance of LLM-based systems with RAG.
- Advisor: Prof. Dr. Ismayle S. Santos. Co-advisor: Prof. Dr. Rossana M. C. Andrade.

MSc in Computer Science

State University of Ceará (UECE)

Mar. 2024 – Mar. 2026

- Dissertation: testing Big Data systems and data quality, combining a multivocal literature review with an LLM/RAG-assisted support tool.

Postgraduate Specialisation in Full Stack Development (360h)

Centro Universitário União das Américas — Descomplica

2023 – 2024

BSc in Computer Engineering

Federal Institute of Ceará (IFCE)

2017 – 2023

CERTIFICATIONS

ISTQB® Certified Tester — Foundation Level (CTFL)

BSTQB®

Mar. 2025

RESEARCH INTERESTS

Testing LLM/RAG systems · Semantic oracles · Data quality · Big Data testing · Multivocal reviews (MLR/MSR) · CI/CD and semantic regression · SE4AI

Junior QA Analyst

PinkApp

2021 – 2022 Remote

- Web test automation with Selenium, load testing with JMeter/Taurus and manual testing within an agile process.

QA Intern

Bynd — Technology Services

2019 – 2021 Remote

- Built the company's first automated test suite with Appium, alongside manual test cycles and defect reporting.

PUBLICATIONS

Journals

[1] Big Data Testing: A Multivocal Literature Review Bridging Academic Research and Industry Practice

Í. S. de Oliveira, I. S. Santos, J. M. de L. Alves, N. N. A. da Silva, R. M. C. Andrade

Journal of Systems and Software (JSS), Elsevier — submitted, under review (2026).

Book Chapters

[2] Big Data Fortaleza Platform: Testing Process

A. K. B. Cavalcante, **Í. S. de Oliveira**, V. T. Oliveira, P. A. M. Oliveira, T. P. Nogueira, I. S. Santos, R. M. C. Andrade

Lecture Notes in Business Information Processing (LNBIP), Springer — extended version of the ICEIS 2025 paper; to appear.

Conferences and Symposia

[3] Experience Report: Test Architecture and Semantic Validation for a RAG System in the Public Sector

Í. S. de Oliveira, F. F. Rodrigues de Oliveira, P. H. M. Maia, I. S. Santos

11th Brazilian Symposium on Systematic and Automated Software Testing (SAST 2026), CBSoft, São Paulo, Brazil. ACM — artefact also accepted in the Artefacts track (Apache 2.0, DOI: 10.5281/zenodo.21837636). In Portuguese.

[4] From Manual Validation to Continuous Quality: An Experience Report on the Evolution of Automated Testing in a Legacy ERP

Í. S. de Oliveira, F. F. Rodrigues de Oliveira, I. S. Santos

14th Workshop on Software Visualization, Evolution and Maintenance (VEM 2026), CBSoft, São Paulo, Brazil — in Portuguese.

[5] "Will I Be Replaced?" Revisited: An Exploratory Replication on AI Tool Perceptions with Software Professionals in Brazil

Í. S. de Oliveira, F. F. Rodrigues de Oliveira, P. H. M. Maia, I. S. Santos

5th Brazilian Workshop on Intelligent Software Engineering (ISE 2026), CBSoft, São Paulo, Brazil. ACM — replication package: DOI 10.5281/zenodo.20947923.

[6] Big Data Fortaleza Platform: Quality Improvement with Testing Process

A. K. B. Cavalcante, **Í. S. de Oliveira**, V. T. Oliveira, P. A. M. Oliveira, T. P. Nogueira, I. S. Santos, R. M. C. Andrade

Proceedings of the 27th International Conference on Enterprise Information Systems (ICEIS 2025), vol. 2, pp. 227–236, SciTePress — DOI: 10.5220/0013290500003929.

[7] Quality of Big Data Systems: a Systematic Review of Practices, Methods and Tools

Í. S. de Oliveira, J. M. de L. Alves, S. C. Alcântara, I. S. Santos, R. M. C. Andrade

Proceedings of the 23rd Brazilian Symposium on Software Quality (SBQS 2024), pp. 22–31, SBC

TECHNICAL SKILLS

Testing & Quality

Playwright · Cypress · Selenium · Appium · Pytest · JUnit 5 · TIR (Protheus) · Allure · TestLink · Postman

Performance & Security

k6 · JMeter · Locust · Taurus · OWASP ZAP

CI/CD & Infrastructure

Jenkins · GitLab CI · Azure DevOps · Docker · Podman · AWS (EC2) · Gitflow

Data & AI

PySpark · spaCy · LLMs and RAG · Embeddings · pandas · LDA · DuckDB

Languages & Frameworks

Python · JavaScript/TypeScript · Java · Go · React · React Native · Spring · Django · PostgreSQL

RESEARCH ARTEFACTS

Test architecture for a RAG system
Open reference implementation (Apache 2.0). DOI: 10.5281/zenodo.21837636

Replication package — AI tool perceptions
Instrument, dataset and coding matrix. DOI: 10.5281/zenodo.20947923

ACADEMIC ACTIVITIES

Technical supervision of an undergraduate research fellow (PIBIC)

UECE

2026 – Present

- Project on data and index quality for RAG systems.

LANGUAGES

Portuguese (native) ●●●●●

English ●●●●○

Spanish ●●●○○